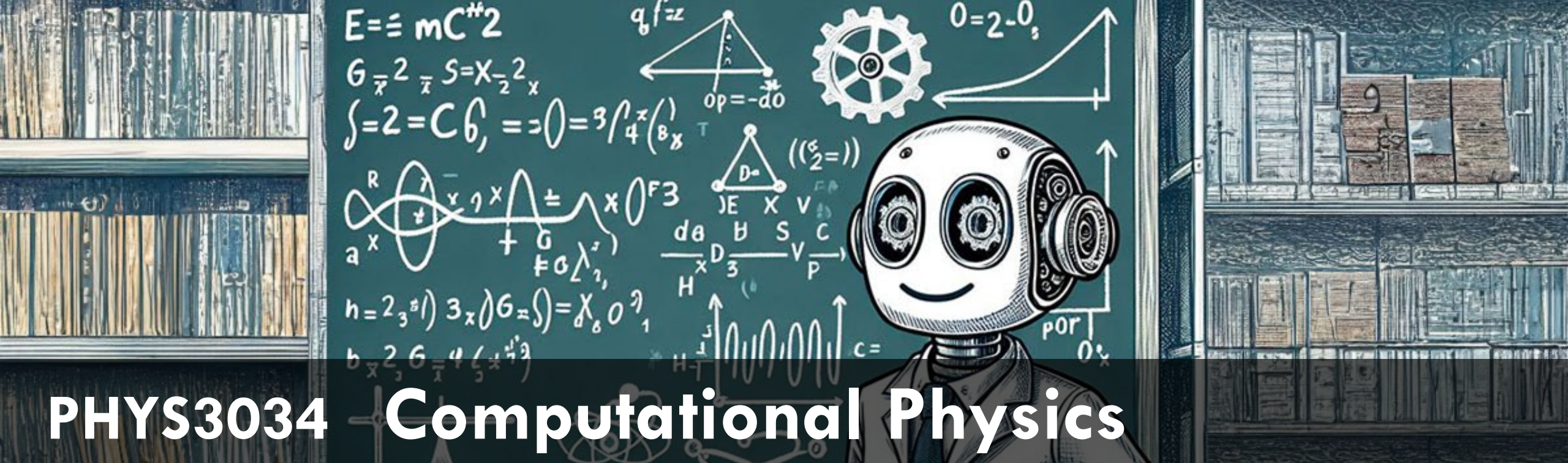




PHYS3034 Computational Physics

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Recap Week 4

- Solved the *diffusion equation* for the heat spike problem using the *forward time centered space* (FTCS) method.
 - $T_i^{n+1} = T_i^n + f (T_{i-1}^n - 2T_i^n + T_{i+1}^n)$, where $f = \kappa\tau/h^2$
 - Works pretty well for diffusion!
- Represented the scheme in matrix form.
 - Multiple time updates become powers of the update matrix, $\mathbf{A}$: $\mathbf{T}^n = \mathbf{A}^{n-1}\mathbf{T}^1$.
- Analyzing the eigenvalues of $\mathbf{A}$ allowed us to understand the behavior of algorithms (including their *stability*) (computer lab):
 - $|\lambda_{\max}|$ tells us about stability, and the sign of eigenvalues allows us to reason about oscillatory behavior.

Lecture 5: Outline

- The *wave* and *advection* equations.
 - Describe propagation of an amplitude profile.
- The Forward Time Centred Space (FTCS) method for advection.
 - *Unstable for all time steps τ !*
- *von Neumann analysis* for assessing algorithm stability.
- *Lax method* for advection.
 - Stabilizes by adding diffusion.
 - Courant-Friedrichs-Lewy (CFL) condition for stability.
- Can yield *worse* performance with low τ .



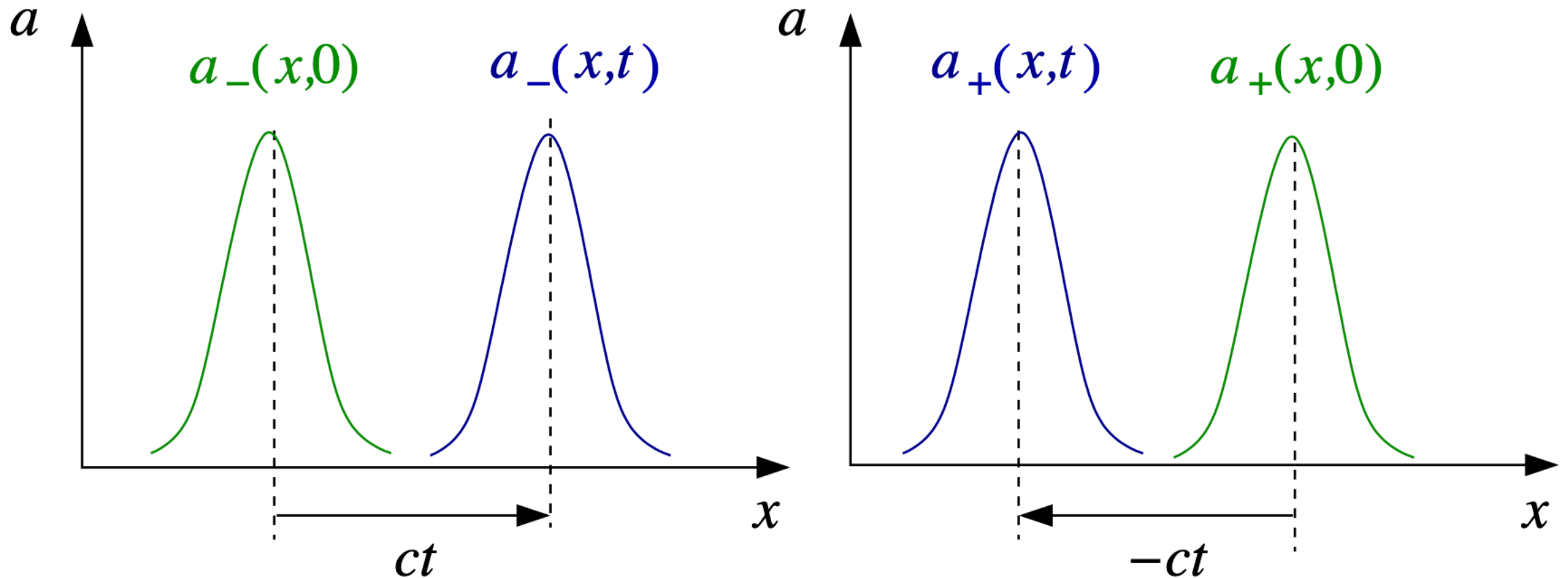
The Wave and Advection Equations

The Wave Equation

- $\frac{\partial^2 a}{\partial t^2} = c^2 \frac{\partial^2 a}{\partial x^2}.$
 - Position x , time t , constant c .
 - Describes the *propagation of a disturbance* in x at speed c .
- Initial Condition: $f(x) = a_{\pm}(x, 0).$
- There are two basic analytic solutions: $a_{\pm}(x, t) = f(x \pm ct)$
 - for some arbitrary function, f .
-  **Exercise** (*easy*): Show that these two solutions satisfy the PDE.

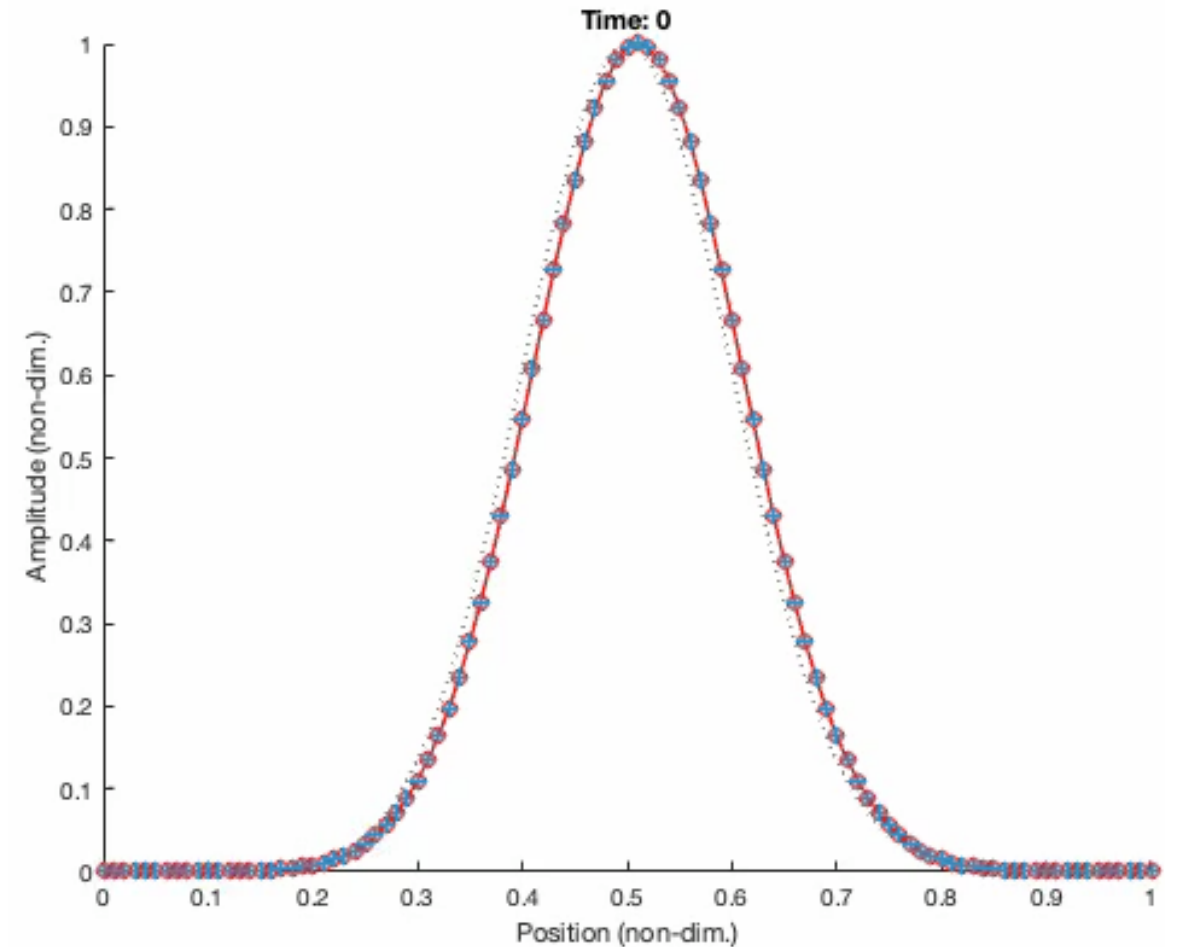
Over time, the initial profile *shifts* to the left/right (+/−) in x :

$$a_{\pm}(x, t) = f(x \pm ct).$$



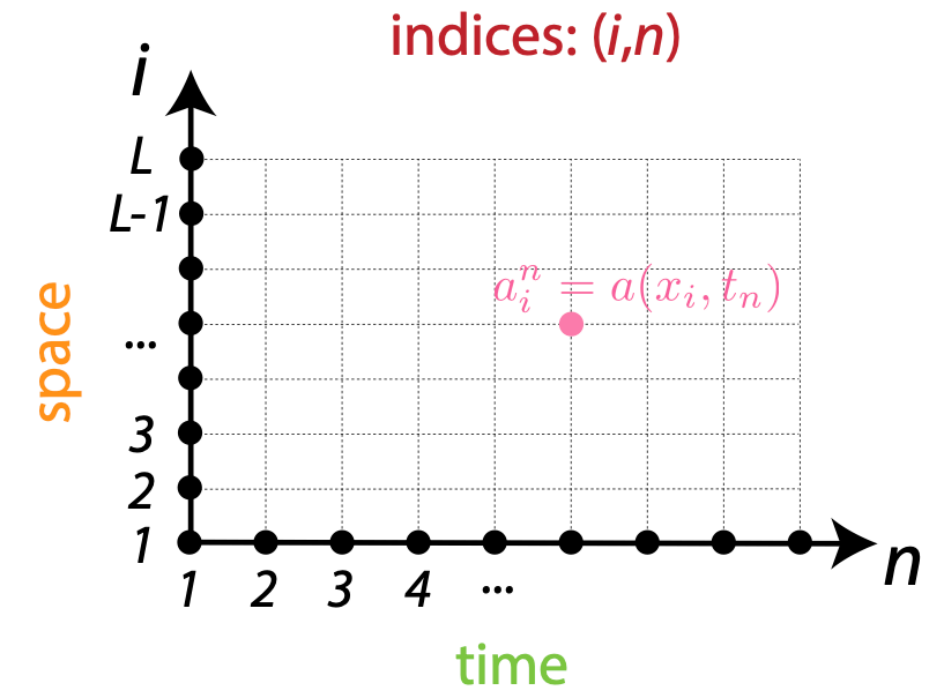
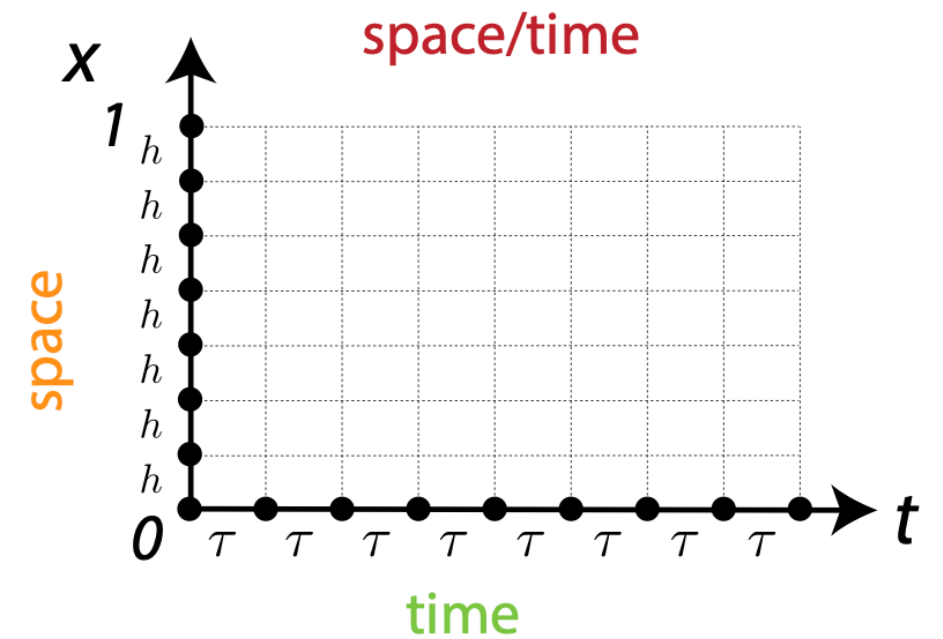
The Advection Equations

- The wave equation can be factorized as
$$\left(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\right)a = 0.$$
- These two operators define the *linear advection equations* (wave speed $c > 0$):
 - $\frac{\partial a_+}{\partial t} = c\frac{\partial a_+}{\partial x}$ with solution: $a_+ = f(x + ct)$.
 - $\frac{\partial a_-}{\partial t} = -c\frac{\partial a_-}{\partial x}$ with solution: $a_- = f(x - ct)$.
- *Simpler* since each equation describes propagation in one direction
 - We'll focus on $\frac{\partial a}{\partial t} = -c\frac{\partial a}{\partial x}$ 😊
 - ? Is this a left-moving or right-moving wave?
- *FYI*: We could easily 'remove' c by rescaling time/space with the choice $t_s = L_s/c$, but let's keep it 😊



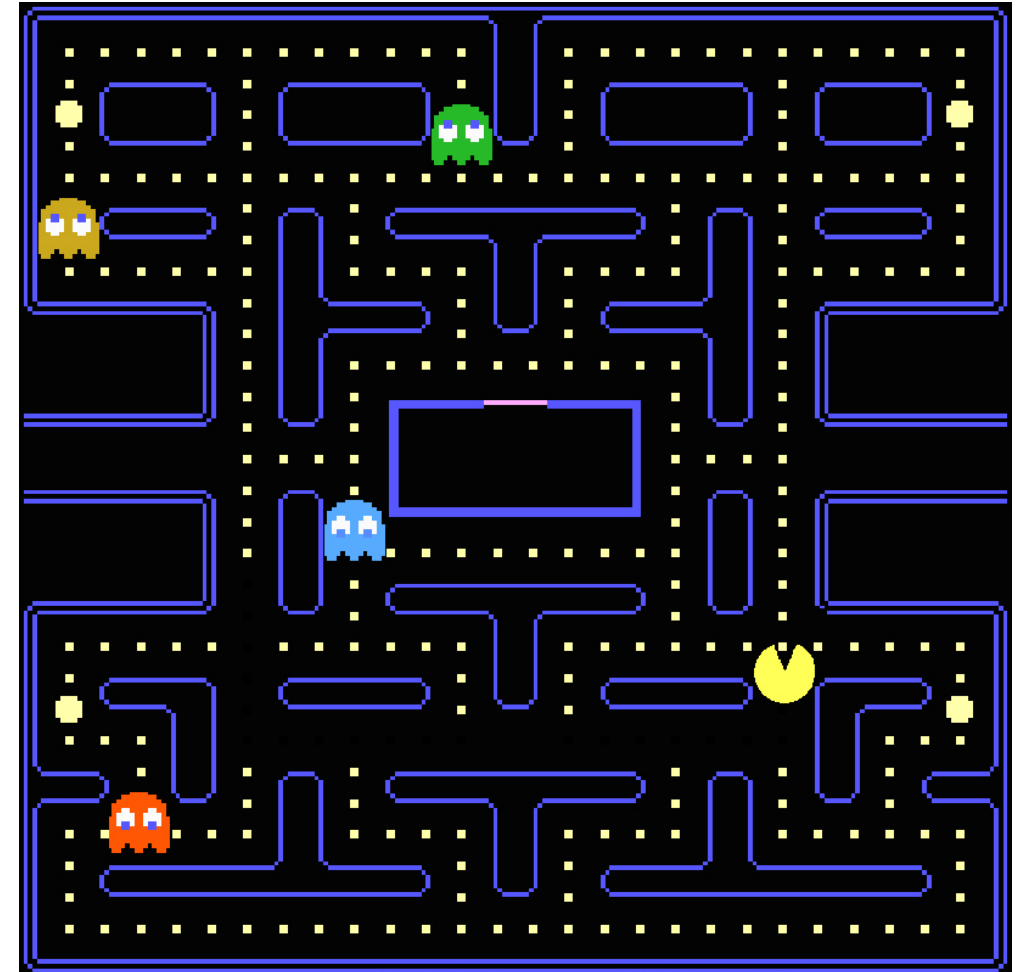
Reminder: discretising space and time

- *Reminder* (from last week): We solve on a *discrete grid* in space and time:
 - $x_i = (i - 1)h$ with $h = (L - 1)^{-1}$ for $i = 1, 2, \dots, L$.
 - $t_n = (n - 1)\tau$ with $n = 1, 2, \dots$ (so $t_1 = 0$).
 - Using the notation: $a_i^n = a(x_i, t_n)$.



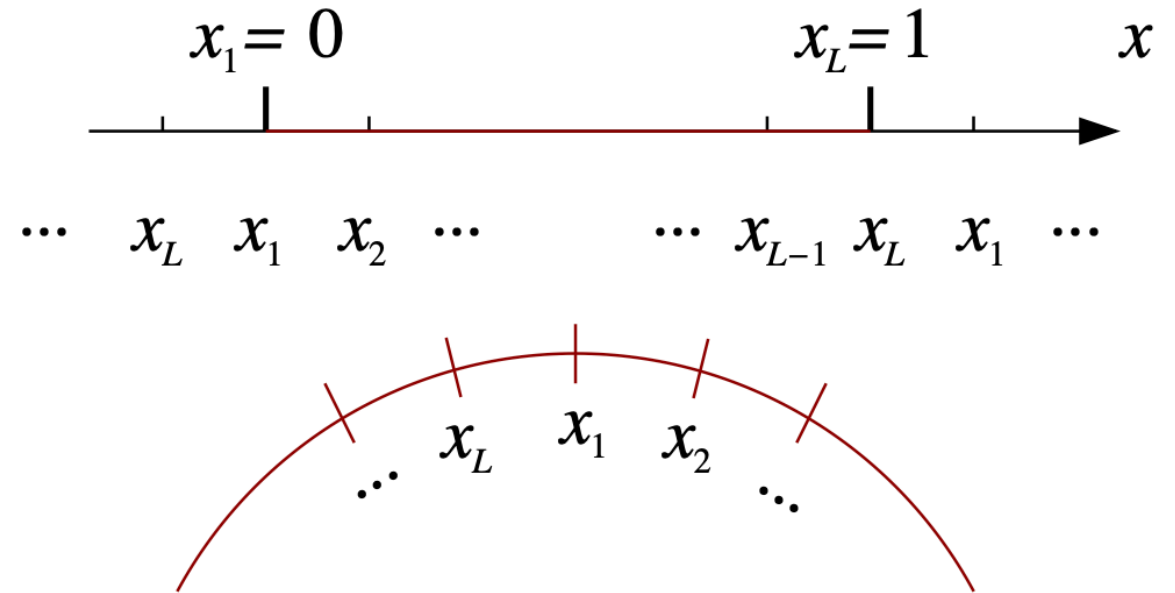
Periodic Boundary Conditions

- (Remember *Dirichlet*?: specify the value at the boundary and solve inside).
- Another choice is *periodic boundary conditions*
- We assume we're on a circular strip of paper, such that walking off the right end takes us to the left end.
 - Just like walking out the right/left tunnels in Pacman.



Periodic Boundary Conditions

- Assume one step to the right of $x_L = 1$ coincides with $x_1 = 0$.
- Assume one step to the left of $x_1 = 0$ coincides with $x_L = 1$.
- Mathematically, this corresponds to the *two conditions* (for all t_n):
 1. $a_{L+1}^n = a_1^n$, and
 2. $a_0^n = a_L^n$.



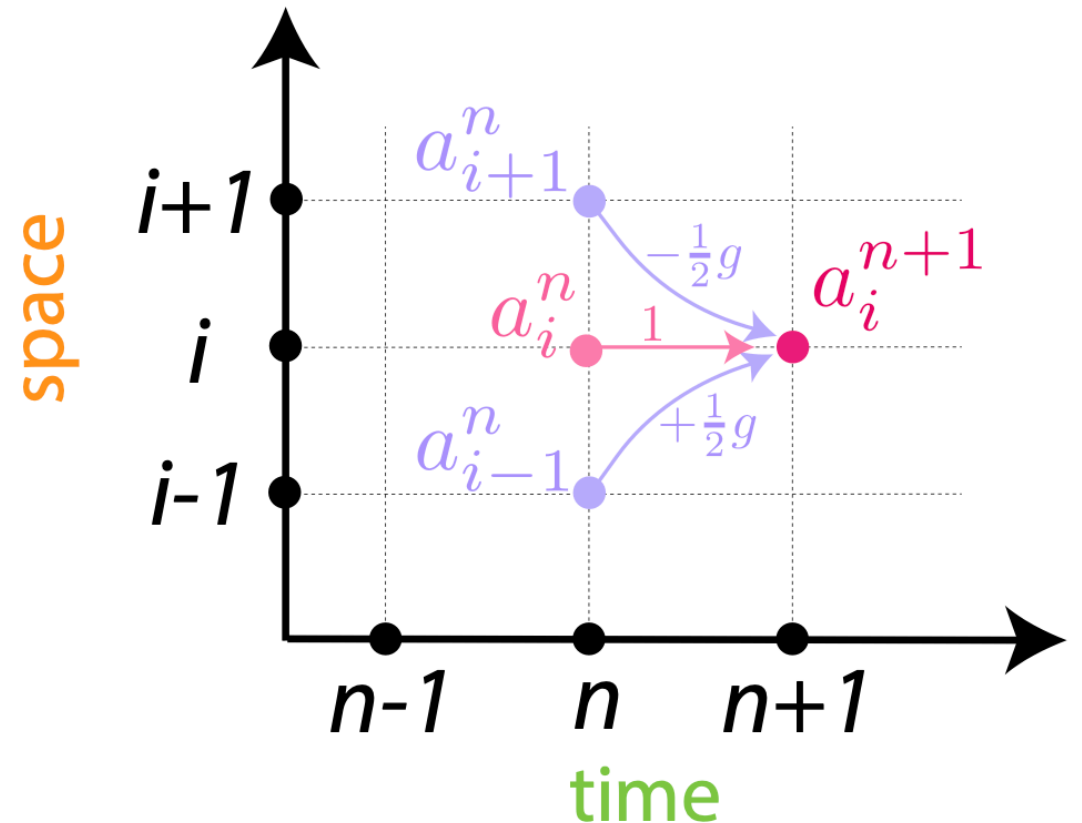
FTCS for advection

The Forward Time Centred Space (FTCS) Algorithm for Advection, $\frac{\partial a}{\partial t} = -c \frac{\partial a}{\partial x}$

- *Forward difference* approximation for $\frac{\partial a}{\partial t}$ ('step forward in time'):
 - $\frac{a_i^{n+1} - a_i^n}{\tau} + O(\tau).$
- *Centered difference* approximation for $\frac{\partial a}{\partial x}$ ('a step forward and backward in space'):
 - $\frac{a_{i+1}^n - a_{i-1}^n}{2h} + O(h^2).$
- Yields: $a_i^{n+1} = a_i^n - \frac{1}{2}g (a_{i+1}^n - a_{i-1}^n).$
 - Constants all wrapped up in $g = \frac{c\tau}{h}.$

The FTCS method for advection

$$a_i^{n+1} = a_i^n - \frac{1}{2}g (a_{i+1}^n - a_{i-1}^n)$$



All the updates $a_i^{n+1} = a_i^n - \frac{1}{2} g(a_{i+1}^n - a_{i-1}^n)$

$$i = 1 : \quad a_1^{n+1} = a_1^n - \frac{1}{2} g \left(a_2^n \boxed{-a_L^n} \right)$$

$$i = 2 : \quad a_2^{n+1} = a_2^n - \frac{1}{2} g \left(a_3^n - a_1^n \right)$$

$$\vdots = \vdots \quad \quad \quad \vdots = \quad \quad \quad \vdots$$

$$i = L - 1 : \quad a_{L-1}^{n+1} = a_{L-1}^n - \frac{1}{2} g \left(a_L^n - a_{L-2}^n \right)$$

$$i = L : \quad a_L^{n+1} = a_L^n - \frac{1}{2} g \left(\boxed{a_1^n} - a_{L-1}^n \right) .$$

- *Periodic BCs*: incorporated in the boxed terms 🙄🙄
- Our a_i^{n+1} are linear combinations of our a_i^n
 - So we can write the updates using *matrix multiplication*.

Matrix version of $a_i^{n+1} = a_i^n - \frac{1}{2}g(a_{i+1}^n - a_{i-1}^n)$

- We can write our updates as $\mathbf{a}^{n+1} = \mathbf{M}\mathbf{a}^n$ with $\mathbf{a}^n = (a_1^n, a_2^n, \dots, a_L^n)^T$ for

- $\mathbf{M} = \mathbf{I} - \frac{1}{2}g\mathbf{D}$ with $\mathbf{D} = \begin{pmatrix} 0 & 1 & 0 & 0 & \dots & 0 & 0 & \boxed{-1} \\ -1 & 0 & 1 & 0 & \dots & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & \dots & 0 & 0 & 0 \\ \vdots & & & & & & & \vdots \\ 0 & 0 & 0 & 0 & \dots & -1 & 0 & 1 \\ \boxed{1} & 0 & 0 & 0 & \dots & 0 & -1 & 0 \end{pmatrix}$

- *Periodic BCs*: come from the boxed elements 👁👁

In code, advection.ipynb: Once we have constructed $\mathbf{M}$, a step forward in time is:
`amp = M @ amp;` 👁👁

FCTS solution for a Gaussian Initial Value Problem

- *Initial profile:* A cute little Gaussian 🧒 centered at $x = \frac{1}{2}$:
 - $a(x, 0) = \exp\left[-\frac{1}{2\sigma^2} \left(x - \frac{1}{2}\right)^2\right]$.
- The analytic solution (with periodic BCs) is:
 - $a(x, t) = \exp\left\{-\left[(x - ct \bmod 1 + h) - \frac{1}{2}\right]^2 / 2\sigma^2\right\}$.
 - Translation of the pulse in x by ct (with periodic BCs via the mod).

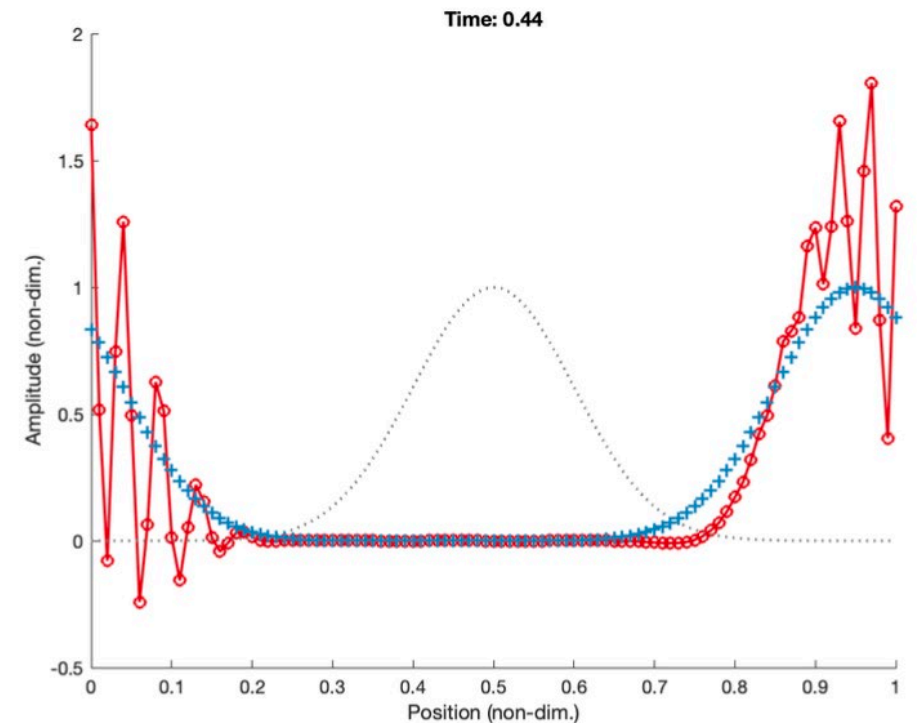
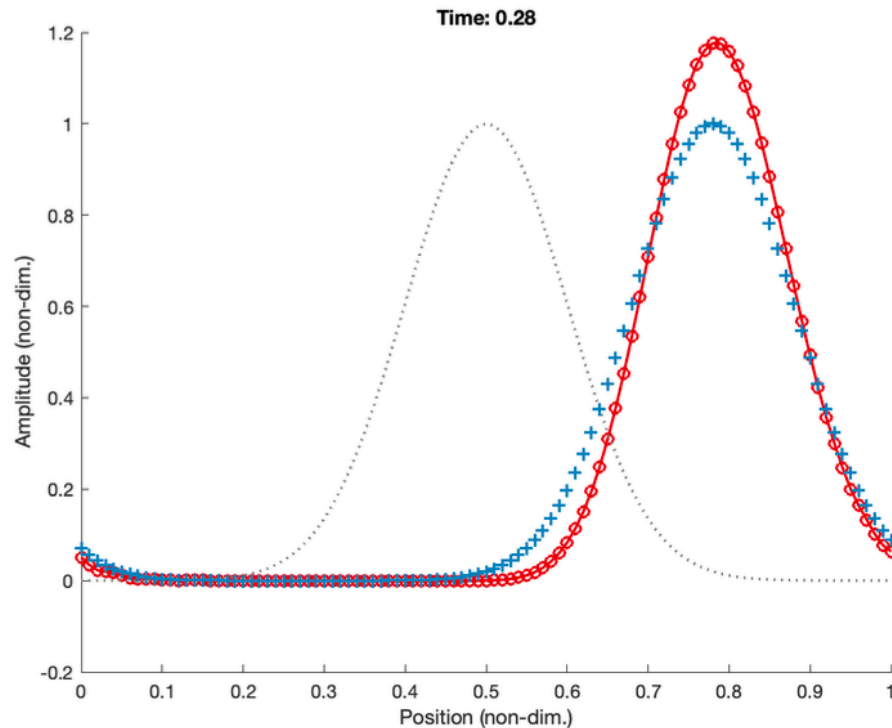
FTCS for advection:

advection.ipynb with `what_method = 'ftcs'`

- *Remember* (last week): For 1D *diffusion*, FTCS is stable for $\tau < h^2/2\kappa$.
- Let's investigate using `advection.ipynb`, setting `what_method = 'ftcs'`.
 - Implements FTCS for the Gaussian pulse IVP.
 - Uses the `construct_update_matrix` function to generate the FTCS update matrix for advection (later we will switch to a different method).
 - Defaults are $\tau = h = 0.01$ and $c = 1$ (so $g = \frac{c\tau}{h} = 1$).
 - Total number of time steps, $N_{steps} = (1 + h)/c\tau$.
 - Total distance of propagation: $N_{steps} \times c\tau = 1 + h$ the size of the domain, so the pulse should *return to its starting position*
 - Animates the numerical and analytic solutions.
 - Also displays as a $a(x, t)$ surface over the (x, t) plane.

FTCS is unstable for all choices of τ 🙄

- The amplitude grows at each time step.
- Doesn't improve with smaller τ : just *delays the inevitable*.
- Matrix stability: spectral radius $\rho(\mathbf{M}) > 1$ *for all* τ ! 🙄



Von Neumann analysis

Von Neumann Analysis

- *von Neumann analysis* is more useful in practice than (cumbersome) Matrix stability analysis 😞
 - von Neumann (1903–1957): 20th Century Hungarian mathematician, physicist, computer scientist, engineer, ...
- Involves *looking for complex wave solutions* 👁️ (we saw these driving the advection instability above):
 - $a(x, t) = A(t)e^{ikx}$
 - *wavenumber* k (spatial wavelength $\lambda = 2\pi/k$).
 - complex time-dependent *amplitude* $A(t)$.
 - (🔥 these don't come from nowhere: a Fourier basis expansion)
- ★ If $|A(t)|$ grows with time, for any k , *the method is unstable* ★

Von Neumann Analysis

- We use the discrete version of von Neumann's complex wave solutions, $a(x, t) = A(t)e^{ikx}$, as:
 - $a_j^n = a(x_j, t_n) = A^n e^{ikjh}$
 - **NB:** here *we use j for the spatial index* to avoid confusion with $i = \sqrt{-1}$.
- Then $a_j^{n+1} = A^{n+1} e^{ikjh} = \xi A^n e^{ikjh}$, where we have defined a complex *amplification factor*, ξ , as how the amplitude, A , of some complex wave solution changes in a step $n \rightarrow n + 1$:
 - $\xi = \frac{A^{n+1}}{A^n}$.
 - This is important because an unstable method grows in amplitude over successive iterations...
- ★ *If $|\xi| > 1$ for any k in a time step the method is unstable*
 - NB: $|\xi|^2 = \text{Re}(\xi)^2 + \text{Im}(\xi)^2$.
- Our strategy in *von Neumann stability analysis* is to insert our trial oscillatory solution into the numerical scheme and see how ξ depends on τ and h ! 😊

Von Neuman to FTCS for advection

$$a_j^{n+1} = a_j^n - \frac{1}{2}g \left(a_{j+1}^n - a_{j-1}^n \right), \text{ with } g = \frac{c\tau}{h}$$

- ? Consider a trial solution, $a_j^n = A^n e^{ikjh}$
and compute $\xi = \frac{A^{n+1}}{A^n}$.

Von Neuman to FTCS for advection

$$a_j^{n+1} = a_j^n - \frac{1}{2}g \left(a_{j+1}^n - a_{j-1}^n \right), \text{ with } g = \frac{c\tau}{h}$$

- ? Consider a trial solution, $a_j^n = A^n e^{ikjh}$
and compute $\xi = \frac{A^{n+1}}{A^n}$.
- 👁️ $\xi A^n e^{ikjh} = A^n e^{ikjh} - \dots$
 - $\dots \frac{1}{2}g \left[A^n e^{ik(j+1)h} - A^n e^{ik(j-1)h} \right]$
- i.e., $\xi = 1 - \frac{1}{2}g (e^{ikh} - e^{-ikh})$
- So $\xi = 1 - ig \sin(kh)$
 - using $\sin(kh) = \frac{1}{2i} (e^{ikh} - e^{-ikh})$
- ? Determine $|\xi|$ and assess stability.

Von Neuman to FTCS for advection

$$a_j^{n+1} = a_j^n - \frac{1}{2}g \left(a_{j+1}^n - a_{j-1}^n \right), \text{ with } g = \frac{c\tau}{h}$$

- ? Consider a trial solution, $a_j^n = A^n e^{ikjh}$
and compute $\xi = \frac{A^{n+1}}{A^n}$.
- 👁️ $\xi A^n e^{ikjh} = A^n e^{ikjh} - \dots$
 - $\dots \frac{1}{2}g \left[A^n e^{ik(j+1)h} - A^n e^{ik(j-1)h} \right]$
- i.e., $\xi = 1 - \frac{1}{2}g (e^{ikh} - e^{-ikh})$
- So $\xi = 1 - ig \sin(kh)$
 - using $\sin(kh) = \frac{1}{2i} (e^{ikh} - e^{-ikh})$

- ? Determine $|\xi|$ and assess stability.
- 👁️ $|\xi| = \sqrt{1 + g^2 \sin^2(kh)}$
- 👁️ Since $g^2 \geq 0$ and $\sin^2(kh) \geq 0$,
 $\Rightarrow |\xi| > 1$ for some k *for all* τ .
- 👁️ So FTCS is *unconditionally unstable*
(for all values of τ): it is a 'useless'
numerical method 😞

Fixing FTCS for advection

- We need to 'fix' the inherent instability.
- Peter Lax  introduced a *simple fix* for this inherent instability 
- ? Which do you think it was?:

$$1. a_j^{n+1} = a_j^n - \frac{1}{2}g \left(\boxed{2a_{j+1}^n} - a_{j-1}^n \right)$$

$$2. a_j^{n+1} = \boxed{\frac{1}{2} \left(a_{j-1}^n + a_{j+1}^n \right)} - \frac{1}{2}g \left(a_{j+1}^n - a_{j-1}^n \right)$$

$$3. a_j^{n+1} = \boxed{\frac{1}{3} \left(a_{j-1}^n + a_j^n + a_{j+1}^n \right)} - \frac{1}{2}g \left(a_{j+1}^n - a_{j-1}^n \right)$$

Lax method

The Lax method

- Lax's idea was to introduce some 'artificial diffusion' (averaging) to counteract the FTCS instability.
- Replace a_i^n term on RHS in FTCS by a *spatial average*.
- *The Lax method for advection:*
 - $$a_j^{n+1} = \boxed{\frac{1}{2} (a_{j-1}^n + a_{j+1}^n)} - \frac{1}{2} g (a_{j+1}^n - a_{j-1}^n)$$
 - $g = c\tau/h$
- Compare to *FTCS*:
$$a_j^{n+1} = \boxed{a_j^n} - \frac{1}{2} g (a_{j+1}^n - a_{j-1}^n).$$



Lax method for advection: von Neumann analysis

- We can compute amplification factor $\xi = \cos(kh) - ig \sin(kh)$
 - So $|\xi|^2 = \cos^2(kh) + g^2 \sin^2(kh) = 1 + (g^2 - 1) \sin^2(kh)$
- Hence $|\xi|^2 \leq 1$ for $g^2 \leq 1$, i.e., stable for $g \leq 1$.
- $g \leq 1$ corresponds to $\tau \leq \tau_{\max} = \frac{h}{c}$.
 - This is the *Courant-Friedrichs-Lewy (CFL) condition*.
- Physically, $\tau_{\max} = h/c$ corresponds to the propagation of one spatial step per time step.

Lax method for advection: Matrix version

- $\mathbf{a}^{n+1} = (\mathbf{A} - \frac{1}{2}g\mathbf{D}) \mathbf{a}^n,$
 - where $\mathbf{D}$ is defined as before but we just slightly tinker with $\mathbf{A}$ to get our averaging in:

$$\mathbf{A} = \frac{1}{2}|\mathbf{D}| = \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 & 0 & \dots & 0 & \boxed{1} \\ 1 & 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & 0 & 1 & \dots & 0 & 0 \\ \vdots & & & & & & \vdots \\ 0 & 0 & 0 & 0 & \dots & 0 & 1 \\ \boxed{1} & 0 & 0 & 0 & \dots & 1 & 0 \end{pmatrix}.$$

(Periodic boundary conditions are the boxed terms).

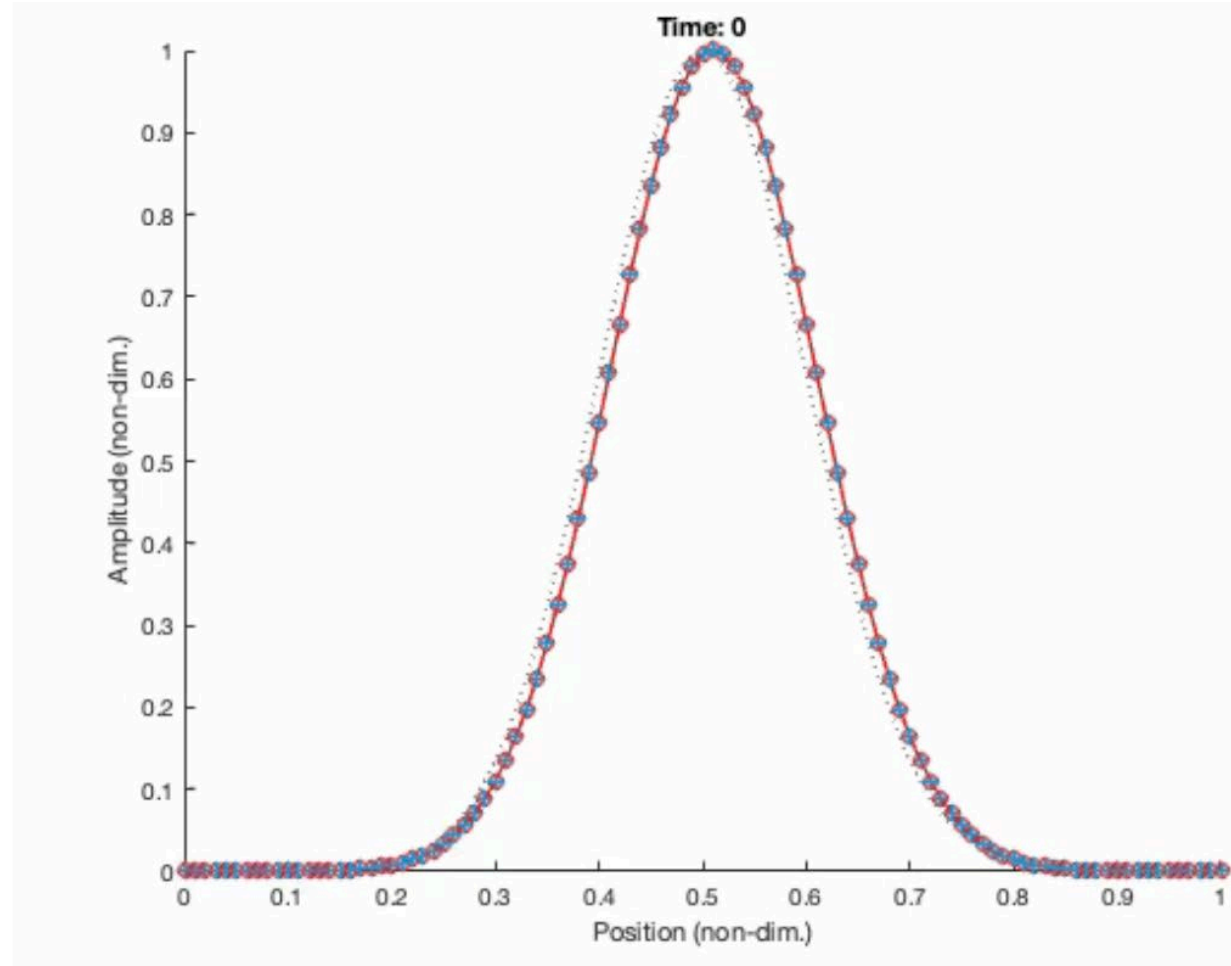
Lax for advection: advection.ipynb with what_method = 'lax'

The *only change* is in how the update matrix is constructed

```
if what_method == 'ftcs':  
  
    # FTCS Update matrix for advection  
    M = np.eye(L) - 0.5*g*D  
  
elif what_method == 'lax':  
  
    # Lax averaging matrix  
    A = 0.5*np.abs(D)  
  
    # Lax Update matrix for advection  
    M = A - 0.5*g*D  
  
else:
```

Lax method

- 😓 For $\tau > \tau_{\max}$: *unstable*.
- 😂 For $\tau = \tau_{\max} = \frac{h}{c}$: *exact* propagation!
- For $\tau < \tau_{\max}$: pulse spreads (*diffuses*) as it propagates.
- Smaller time steps τ don't always yield more accurate solutions 😲



Good Luck out there!

- This course is a *very brief introductory sketch* into some of the principles of using computers to simulate the solution of physical equations.
- In the next few years when you've flown from this here nest 🐦, you might get your dirty hands on some gnarly equations. *Then what do you do?*

1. **The same principles apply**

- Similar methods can often be adapted to different problems.

2. **Look to the literature for guidance (textbooks and refereed journal articles)**

- This course focused on *finite differencing* methods, but there are many other powerful approaches (e.g., finite elements, spectral methods, ...)

3. **Always try the simplest thing first**

- Solve a simpler problem/a limited case first, then build up if required.

4. **Always test any numerical method on known analytic solutions to understand/validate its behavior.**

Computational Physics Cheat Sheet

Numerical errors

Floating point versus truncation errors

Global versus local error $E_g \approx O(\tau^{k-1})$

Constructing algorithms

1st derivative: forward difference approximation:

$$f'(t) = \frac{f(t+\tau) - f(t)}{\tau} + O(\tau)$$

1st derivative: centered difference approximation:

$$f'(t) = \frac{f(t+\tau) - f(t-\tau)}{2\tau} + O(\tau^2)$$

2nd derivative: centered difference approximation:

$$f''(t) = \frac{f(t+\tau) - 2f(t) + f(t-\tau)}{\tau^2} + O(\tau^2)$$

$$\text{1D Diffusion} \quad \frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial x^2}$$

$$\text{FTCS: } T_i^{n+1} = T_i^n + f(T_{i-1}^n - 2T_i^n + T_{i+1}^n) \quad f = \frac{\kappa\tau}{h^2}$$

$$\text{Matrix form: } \mathbf{T}^{n+1} = (\mathbf{I} + \mathbf{D})\mathbf{T}^n = \mathbf{A}\mathbf{T}^n \quad \text{Stable: } \tau \leq \frac{h^2}{2\kappa}$$

$$\text{Advection} \quad \frac{\partial a}{\partial t} = -c \frac{\partial a}{\partial x}$$

$$\text{FTCS: } a_i^{n+1} = a_i^n - \frac{1}{2}g(a_{i+1}^n - a_{i-1}^n) \quad g = \frac{c\tau}{h}$$

$$\text{Matrix form: } \mathbf{a}^{n+1} = (\mathbf{I} - \frac{1}{2}g\mathbf{D})\mathbf{a}^n \quad \text{Stability: (unstable)}$$

$$\text{Lax: } a_j^{n+1} = \frac{1}{2}(a_{j-1}^n + a_{j+1}^n) - \frac{1}{2}g(a_{j+1}^n - a_{j-1}^n)$$

$$\text{Stability: } g \leq 1$$

Non-dimensionlization

Introduce new dependent and independent variables by rescaling each by chosen constants.

Do not rescale constants/parameters.

$$\bar{\mathbf{r}} = \frac{\mathbf{r}}{L_s}, \quad \bar{t} = \frac{t}{t_s}, \quad \bar{\mathbf{v}} = \frac{\mathbf{v}}{L_s/t_s}.$$

Algorithms for ODEs

$$\text{Dynamics:} \quad \frac{d\mathbf{r}}{dt} = \mathbf{v}, \quad \frac{d\mathbf{v}}{dt} = \mathbf{a}(\mathbf{r}, \mathbf{v}, t).$$

$$\text{Euler:} \quad \begin{aligned} \mathbf{r}_{n+1} &= \mathbf{r}_n + \tau \mathbf{v}_n, \\ \mathbf{v}_{n+1} &= \mathbf{v}_n + \tau \mathbf{a}_n. \end{aligned}$$

$$\text{General form:} \quad \frac{d\mathbf{x}}{dt} = \mathbf{f}[\mathbf{x}(t), t]$$

$$\text{Euler (RK1):} \quad \mathbf{x}(t+\tau) = \mathbf{x}(t) + \tau \mathbf{f}^{(1)}$$

$$\text{RK4:} \quad \mathbf{x}(t+\tau) = \mathbf{x}(t) + \frac{1}{6}\tau \left[\mathbf{f}^{(1)} + 2\mathbf{f}^{(2)} + 2\mathbf{f}^{(3)} + \mathbf{f}^{(4)} \right]$$

Stability analysis

Matrix power formulation: spectral radius $\rho(\mathbf{A}) \leq 1$

von Neumann: trial solution: $a(x, t) = A(t)e^{ikx}$

$$\text{Amplification factor: } \xi = \frac{A^{n+1}}{A^n} \quad \text{Stability: } |\xi| \leq 1$$